



CSC1024 Final Project - Python Powered Social Media Content Planner

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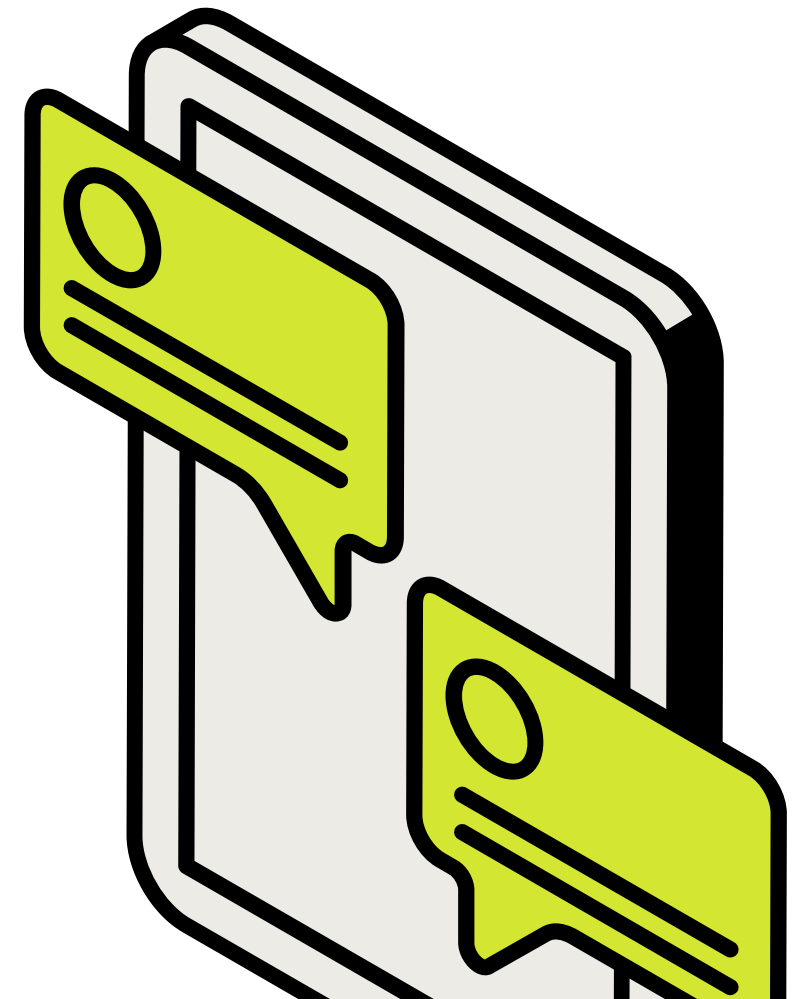
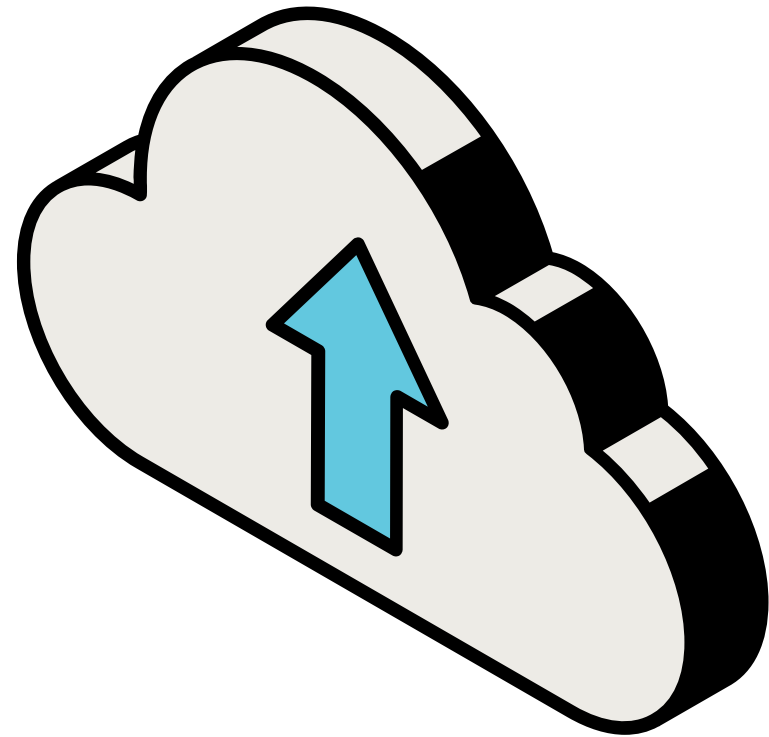
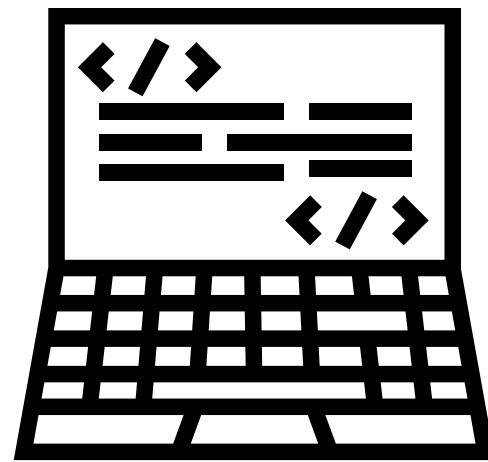


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01

KEY FEATURES & DESIGN DECISIONS



Introduction

Overview:

A command-line interface (CLI) application built in Python for content creators, social media managers, and strategists. Simplifies content management through post planning, lifecycle tracking, and analytics reporting.

Core Objectives:

Organize Content: Plan ideas and schedule posts across multiple platforms (Instagram, TikTok, X).

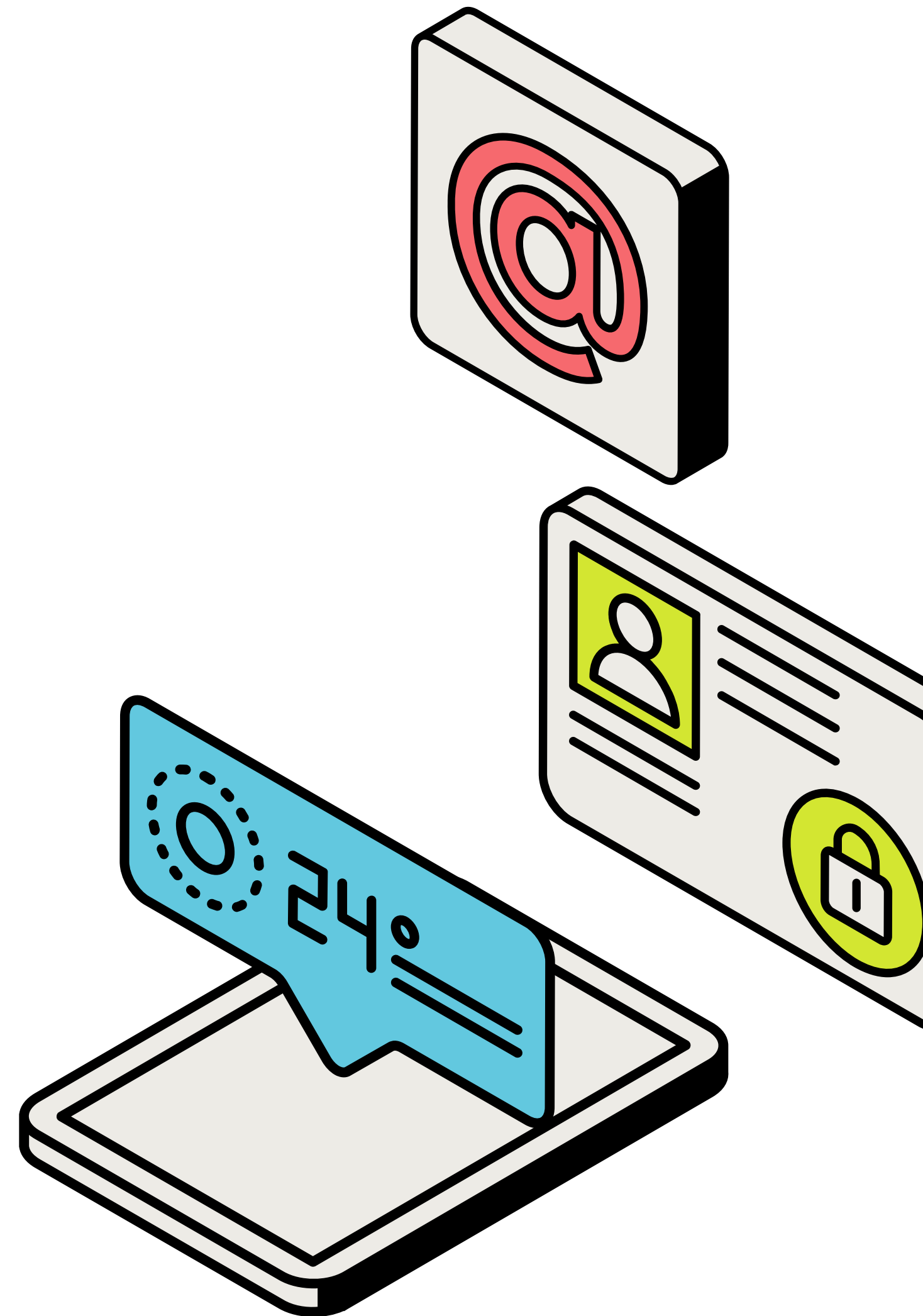
Track Status: Monitor post progress through distinct lifecycle stages (Draft → Scheduled → Posted).

Analyze Performance: Record post engagement and identify top-performing platforms and posts.

Technical Highlights:

Built using pure Python standard libraries (os, datetime, unittest) with zero third-party dependencies.

Persistent storage powered by lightweight flat-file handling.



Functions

Add a New Post Idea

Update Post Ideas

Record Engagement Metrics



Functions

Display Content Calendar

Delete Post Ideas

**Generate Performance
Report**



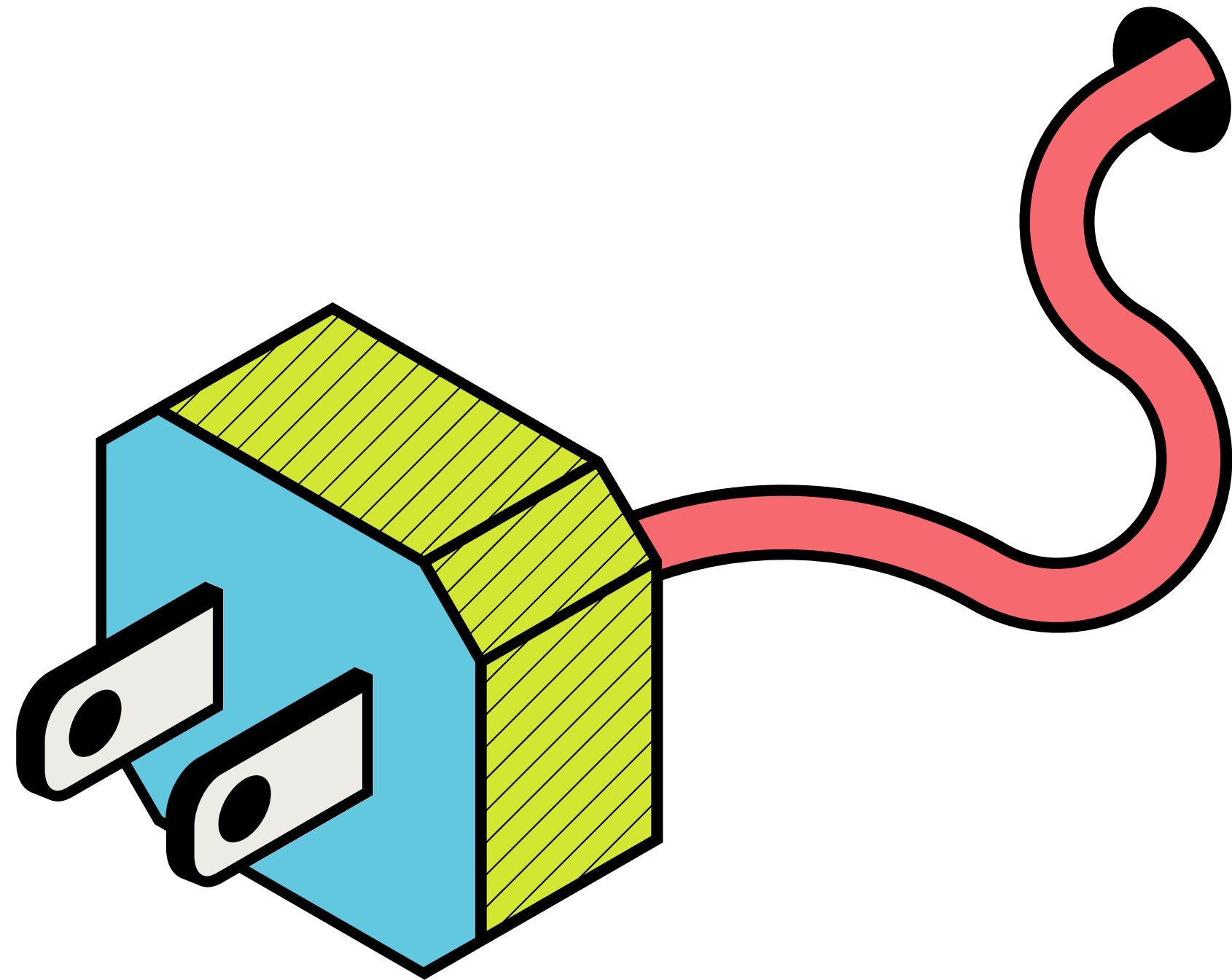
Workflow

User selects function from main menu.

Function is executed.

Data is read/written into file.

Returns user to main menu again and repeats.



Key Features

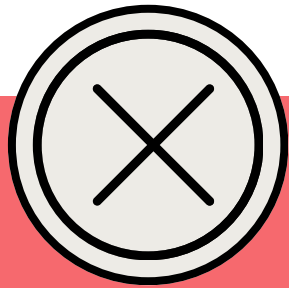


Use of text Files

- **Easier to handle data**
- **Easy to debug**
- **More suitable for a small system**
- **Data persist after program termination**



Key Features

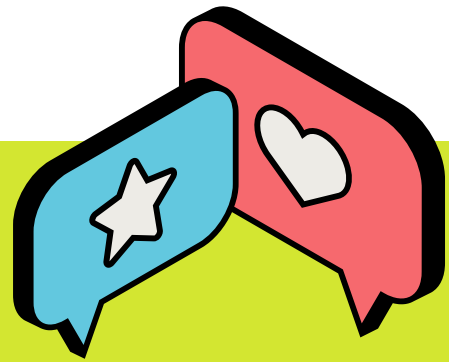


Data Validation for date function

- Splits the input using dash as the date separator
- Check the input contains exactly 3 parts (y-m-d)
- Validates format as YYYY-MM-DD by checking :
 - Year has 4 digits
 - Months has 2 digits
 - Days has 2 digits
- Ensures all parts contains numerical only using is digit



Key Features



Menu System

- User friendly
- Easy to implement



Design Decisions

Custom In-Memory Selection Sort Algorithm:

Organizes content calendar entries chronologically by date without relying on external sorting modules or libraries.

Pipe-Delimited (|) Flat-File Schema:

Standardized format across platforms.txt, posts.txt, and engagement.txt for fast read/write operations and ease of maintenance.

Referential Integrity & Cascading Delete:

Prevents orphan engagement data by automatically deleting corresponding metrics when a post is deleted.

Defensive Input Handling:

Strips reserved delimiter characters (|) from user captions and enforces non-negative integer validation for engagement statistics.



02

System Demonstration



Thank you.

Presentation Link:

https://drive.google.com/file/d/1iFl6tTu0qJECWvk3J6ic1xcC_rzr1cEL/view?usp=sharing

